



$$-(r_j - r_i)^T M (r_j - r_i)$$

geo.pos3d

$$1[j \text{ ancestor of } i]$$

kin.ancestor

$$\text{flow}^+(\text{support})(i, j)$$

ix.force_flow

$$g_{\text{task}} \text{next}(i, j)$$

task.next_contact

$$\text{logit}(i, j) = \frac{\langle q_i, k_j \rangle}{\sqrt{d}} + b_{\text{dist}} + b_{\text{dir}} + b_{\text{kin}} + b_{\text{contact}} + b_{\text{supp/flow}} + b_{\text{held}} + g_{\text{task}} b_{\text{next}} + b_{\text{ui}}$$

learned, data-dependent (no heat)

geo.pos3d

$$\tilde{b}^T (r_j - r_i)$$

ix.contact

contact(i, j)

ix.held_by

held_by(i, j)

ui.label_for

label_for(i, j), above(i, j)

